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March 5, 1965
SUPERSEDES 000-R-96b
in part.

INTERIM FEDERAL SPECIFICATION

RAKE (ASPHALT, ROAD)

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C. 20407, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. - This specification covers asphalt rakes suitable for use in road building, paving work, and various other tarring and graveling tasks, and road rakes suitable for spreading gravel, crush stone, concrete, and other heavy materials.

1.1.1 Federal Specification coverage.- This specification covers only those types of rakes generally purchased by the Federal Government and is not intended to include all of the types that are commercially available.

1.2 Classification

1.2.1 Types. - The hand rakes shall be of the following types, as specified (see 6.1):

Type I. - Asphalt (tar)
Type II - Road

2. APPLICABLE SPECIFICATIONS AND STANDARDS.

2.1 Specifications and standards. - The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

Federal Specification:
NN-H-81- Handles; Ash, for Tools

Federal Standards:
Fed. Std. No. 102 - Preservation, Packaging and packing levels
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424 - Hand Tools, Packaging of

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Military Standard:
MIL-Std-105- Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustrations. - The illustrations shown herein are for the convenience of identification and are not intended to exclude the purchase of rakes which are otherwise in accordance with the requirements of this specification.

3.2 Material. - Materials used in the construction of rakes shall be as specified hereinafter. All materials shall be of uniform quality and free from defects and imperfections that may effect their serviceability or durability.

3.3 Measurements. - Hand rakes shall be measured as illustrated on figure 1.

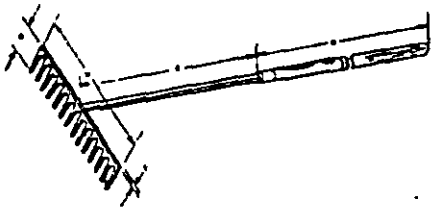


FIGURE 1.—Type I, asphalt (tar) rake.

3.4 Hardness of metal teeth. - The rake teeth shall be of sufficient hardness to comply with the tests specified in 4.4.1 and 4.4.3.

3.5 Heads. - The design of the heads shall be similar to figures 1 and 2. The top and the ends of the head shall be square-edged to semi-round. When tested, the teeth and other parts of the head shall not flatten, chip, crack, deform, or fracture.

3.5.1 Shanks. - The shank shall be centered between ends of the head, plus or minus 1/8 inch, for proper balance. When tested, the shank and head shall not fracture or crack, and the permanent set shall not exceed one inch.

3.6 Handles. - Rake handles shall be in conformance with applicable requirements of MM-H-81. The dimensions shall be as specified in tables I, II, and III. When tested, the handles shall not loosen.

Table I. - Handle Dimensions, types 1 and 2

Diameter adjacent to ferrule		Smallest diameter		Diameter of end (opposite ferrule)	
Min	Max.	Min.	Max	Min.	Max.
Inches 1-3/8	Inches 1-5/8	Inch 1	Inches 1-1/8	Inches 1-3/16	Inches 1-5/16

3.7 Test requirements. - The rakes shall withstand the tests specified in 4.4.1, 4.4.2, 4.4.3, and 4.4.4.

3.8 Nails and rivets. - Nails and rivets where hereinafter specified for attachment of handle or ferrule, shall be steel, oval or round head, not less than 0.115 inch body diameter and not less than 27/32 inch overall length. When any rivet or nail extends entirely through the opposite side of the ferrule, its end shall be neatly headed over the ferrule.

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3.9 Handle coating. - All portions of the handle shall be smoothly finished and shall be adequately covered with two durable coats of transparent lacquer or varnish or one coat after surface treatment utilizing heat, such as flame treating or charring. The coating shall be applied in such a manner to permit visual inspection of the wood and to prevent absorption of water.

3.10 Finish of metal parts. - Heads, ferrules, and exposed shanks shall be coated with paint, enamel, or lacquer.

3.11 Type I, asphalt (tar) rakes - (see 6.1)

3.11.1 Head. - The head shall be formed in one piece from steel, and the shank or any portion thereof shall be either forged integrally with, or welded to the head, at the option of the contractor. Welds, if any, shall form sound substantial joints.

3.11.2 Teeth. - The teeth shall be straight or slightly concaved towards the handle, not to exceed 1/4 inch at the tip in relation to the flat surface of the teeth. The length of the teeth shall not be less than 3-7/8 inches, from the tip to the bottom of the radius. The spacing between the teeth shall not be less than 15/16 inches, when measured 2 inches from the tip.

3.11.3 Ferrule. - A steel tubular ferrule of not less than 8 inches in length shall be provided. The ferrule shall be tapered and fastened to the handle with at least one nail or rivet which shall pass entirely through the handle and shank. If the nail or rivet should penetrate the opposite side of the ferrule, it shall be neatly headed. The seam of the ferrule shall be closed by a continuous weld. The ferrule shall be provided with an attached cap, or the end of the ferrule may be rounded to form a cap.

3.11.4 Shank. - The shank shall have a square or rectangular cross section, any side of which shall be not less than 1/2 inch. The rake shall be of the straight-shank pattern or drop-shank pattern. On the straight-shank pattern rake, the shank shall join the head at the top edge of the head without projecting above the top edge and shall have a slight bend adjacent to the head, as illustrated on figure 1. For the drop-shank pattern rake the shank shall join the head 1/4 to 1 inch below the top edge of the head.

3.11.5 Type I rakes shall be similar to figure I and the dimensions shall conform to those specified in table II.

Table II - Type I, asphalt (tar) rake.

Teeth	Head					Shank		Handle		Test Loads		
	Depth (D)		Thickness (T)	Width (W)		Length (S)		Length (H)				
Number	(Min.)	(Max.)	± 1/32	(Min.)	(Max.)	(Min.)	(Max.)	(Min.)	(Max.)	On handle	On Tooth	On Shan
	Inches	Inches	Inch	Inches	Inches	Inches	Inches	Inches	Inches	pounds	pounds	poun
14	4-3/8	4-5/8	3/8	16	17	18	22	59	61	1000	100	40

3.12 Type 2, road rakes (see 6.1)

3.12.1 Head. - The head and shank shall be formed in one piece from steel.

3.12.2 Teeth. - The teeth shall be straight within 1/8 inch. The length of the teeth shall not be less than 3-1/4 inches, from the tip to the bottom of the radius. The spacings between the teeth shall not be less than 7/8 inches, when measured 2 inches from the tip.

3.12.3 Ferrule. - A steel tubular ferrule not less than 4-1/8 inches long shall be provided. The ferrule shall be tapered and fastened to the handle with at least one nail or rivet. The seam of the ferrule shall be closed by a continuous weld.

3.12.4 Shank. - The rake shall have a single central shank which shall be forced into the handle and reinforced with a ferrule.

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3.12.5 Type 2 rakes shall be similar to figure II, and the dimensions shall conform to those specified in table III

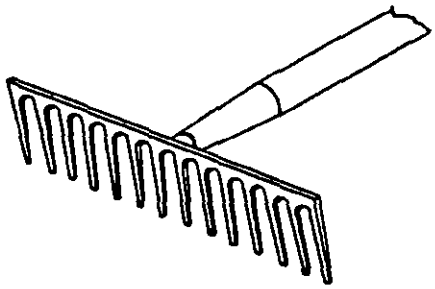


Figure 2, Type 2, road rake

Table III, Type 2, road rake

Head						Handle		Test loads		
Teeth		Depth (D)	Thickness T	Width (W)		Length (H)		On Handle	On Tooth	On Shank
Number	(Min.)	(Max.)	± 1/32	(Min.)	(Max.)	(Min.)	(Max.)			
	Inches	Inches	Inch	Inches	Inches	Inches	Inches	Pounds	Pounds	Pounds
14	3-3/4	4-1/16	3/8	15-1/4	16	65	67	500	100	20
16	3-3/4	4-1/16	3/8	17-1/2	18-1/4	65	67	500	100	20

3.13 Identification marking. - The head or the handle shall be marked in a legible and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determine.

3.14 Workmanship. - The workmanship shall be of the highest grade throughout and in accordance with good commercial practice. Rakes shall be free from any imperfections that may affect their serviceability or durability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection responsibility. - Unless otherwise specified in the contract or purchase order, the supplier is responsibile for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conformed to prescribed requirements.

4.1.1 Component and material inspection. - In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification.

4.2 Sampling procedures. - Sampling procedures shall be in accordance with MIL-Std-105. Data for sampling shall be as stated in table IV.

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Table IV, Sampling Data

Category	Sample Unit	Inspection Level	Acceptable Quality Level	AQL Expressed In Terms of	Reference
Visual examination	1 ea.	II	4.0	Defects per hundred units	Paragraph 4.3.1
Dimensional examination	1 ea.	S-4	1.5	Same	Paragraph 4.3.2
Testing	1 ea	S-4	1.0	Same	Paragraph 4.4

4.3 Examination

4.3.1 Visual examination. - Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking.

4.3.2 Dimensional examination. - Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing. - Each sample unit shall be tested in accordance with 4.4.1, 4.4.2, 4.4.3, and 4.4.4.

4.4.1 Drop test, types 1 and 2. - Teeth and other portions of the head shall not flatten, chip, crack, deform, or fracture when the head is dropped teeth foremost, a distance of 4 feet, three times, on a flat horizontal steel die block. In dropping the head, the free end of the handle shall be loosely pivoted at the level of the surface of the die block. No loosening of the handle shall occur as a result of this test.

4.4.2 Shank test. - The full width of the head and as much of the depth of the head as is possible shall be rigidly clamped or secured in a stationary fixture with the teeth pointed upward and the handle horizontal. The shank and head shall not fracture or crack, and the permanent set shall not exceed 1 inch when the shank test load specified in the applicable table is applied vertically downward to the handle for 120 to 125 seconds at a distance of 36 to 37 inches from the point where the shank joins the head. The vertical distance from the point of application of the load to any flat horizontal surface shall be measured before the load is applied and after it is removed, and difference between these distances shall be the set. No loosening of the handle shall occur as a result of this test.

4.4.3 Tooth strength test. - No tooth shall take a noticeable permanent set after supporting for two minutes, the tooth test load specified for the rake in the applicable table. The head shall be secured or supported in a stationary fixture and the load shall be applied in the center of the tooth in a direction opposite to the handle.

4.4.4 Handle pulling test. - The handle shall not move or loosen when the head is clamped or secured and the handle test load, specified for the rake in the applicable table, is applied for 2 minutes as a pulling load on the handle.

4.5 Inspection of preparation for delivery requirements. - An inspection shall be made to determine that the packaging, packing and marking requirements comply with section 5 of this specification. Defects shall be scored in accordance with table V. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

Table V, Classification of Preparation for Delivery Defects.

Examine	Defects
Marking (exterior and interior)	Omitted: incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

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5. PREPARATION FOR DELIVERY

(For civil agency procurement, Fed. Std. No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Cleaning, preservation, and packaging. - Cleaning, preservation, and packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A.- Cleaning, preservation and packaging shall be in accordance with level A in MIL-H-15424.

5.1.2 Level C.- Rakes shall be preserved and packaged in such a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity for immediate use. This may conform to the supplier's commercial practice, when such meets the requirements of this level.

5.2 Packing. - Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. - Rakes, packaged as specified (see 6.1), shall be packed in accordance with level A of MIL-H-15424.

5.2.2 Level B. - Rakes, packaged as specified (see 6.1), shall be packed in accordance with level B requirements of MIL-H-15424.

5.2.3 Level C.- Rakes, packaged as specified (see 6.1), shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Shipping containers shall be in accordance with rules or regulations of carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil Agencies. - In addition to any special marking required by the contract or order interior and exterior shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-H-15424.

6. NOTES

6.1 Ordering data. - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Type (see 1.2)
(c) Size (number of teeth) (see applicable table)
(d) Level of preservation, packaging and packing required (see section 5)
(e) Marking, if different (see 5.3)

6.3 Transportation description. - The applicable transportation description is as follows:

Types I and II:

Rakes, hand, not wheeled, with iron or steel heads.
Carload weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.
Truckload weight 24,000 pounds.

6.4 Supersession data. - Types I and IV in Federal Specification GGG-R-96b are superseded as follows:

GGG-R-96	This Specification
Type I	Type I
Type IV	Type II

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